Stav Tchizik 316062835
Or Cohen 206631509

Project Proposal

2.1 Research question:

Question: How does the socio-economic status (SES) of a local authority in Israel impact the academic achievements of its students across different subjects and grade levels?

This question is highly specific, measurable, and relies directly on factual government data. It is of high
interest to a defined audience, primarily educational policymakers, municipal leaders, and the Ministry of
Education. Understanding these dynamics can directly inform resource and budget allocations, targeted
educational intervention programs, and systemic reforms aimed at closing societal gaps.

2.2 Dataset

We are utilizing two integrated public datasets from the Israeli Government Data Portal and the Central Bureau of Statistics (CBS):

- Dataset 1: Ministry of Education Academic Scores (aggregated by local authority, grade, and subject).
- Dataset 2: CBS Socio-Economic Index for Local Authorities (2021).

Dataset Characteristics:

- **Size (Points & Features):** In the Ministry of Education Academic Scores dataset, we have 3070 rows, and 12 features. Each row represents authority-grade-subject combinations across years. In the CBS dataset we have 265 rows, each row represents an authority. We only look at the CLUSTER feature in each row.
- **Relevant Columns (Features):** rashut_teur (Authority Name), shichva (Grade level, e.g., 5 or 8), subject_id (Subject, e.g., Math, English), CLUSTER (SES () Cluster 1-10), and INDEX VALUE (Continuous SES rating).
- **Target Column:** score (The average test score for the specific intersection).
- **Important Dataset Information:** The raw data requires a complex merge. The Ministry of Education dataset and the CBS dataset use different numbering systems for local authorities (e.g., Ofakim is '315' in Education data and '31' in CBS data). Data wrangling will be required to create a unified mapping key prior to analysis.

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2.3 Methods

We plan to implement a multi-faceted methodology to thoroughly investigate the data, utilizing several complementary machine learning tools:

1. Exploratory Data Analysis (EDA) & Data Cleaning:
    - Handle missing values and anomalies (e.g., verifying sample sizes in n_students_tested to avoid skewness from small statistical samples).
    - Resolve the local authority ID mismatch by creating a mapping dictionary to perform a clean merge between the educational records and CBS socio-economic data.
    - Plot distributions of test scores across different socio-economic clusters and subjects using visualization libraries.
2. Correlation Analysis: Calculate and visualize correlations between the continuous SES INDEX VALUE and the target score, factoring in grade levels and subjects.
3. Data Splitting: Split the data into Training (50%), Testing (30%), and Validation (20%) sets to prevent overfitting and accurately evaluate model performance.
4. Unsupervised Learning (Clustering): Apply K-Means clustering to the educational performance metrics to identify natural groupings of local authorities based on academic success, and compare how well these academic clusters align with the official CBS socio-economic clusters.
5. Supervised Learning (Regression & Classification):
    - Regression: Build a Random Forest Regressor and a Linear Regression model to predict the continuous score using SES, grade, and subject. Compare their performance metrics.
    - Classification: Convert the score target into a categorical variable (e.g., "High Achievement" vs. "Low Achievement" based on median thresholds). Compare the performance of multiple classifiers (e.g., Random Forest Classifier vs. Logistic Regression).
6. Feature Importance: Extract feature importance scores from the Random Forest models to quantify whether the socio-economic index outweighs other factors (like grade level or specific subjects) in determining academic success.